Lovely Professional University, Phagwara, India

Differential Equation and Vector Calculus (MTH166)
Tutorial sheet - 1

1. Find the order and the degree of the following differential equations. State also whether they are linear or non-linear.

   (a) $y'' + 3y' + 4y = 0.$
   (b) $x^2y'' + xy' + 3y = 5x.$
   (c) $(y')^2 + 3xy' + y = 0.$
   (d) $\sqrt{1+2x^2}\,dx + \sqrt{1+2y^2}\,dy = 0.$
   (e) $\left[1 + (y')^2\right]^{1/2} = x^2 + y.$
   (f) $yy'' + (y')^2 + 4y = \cos t.$
   (g) $(y'')^2 + 3y' + x = 0.$
   (h) $y'' + y' + 5y = \sin x.$
   (i) $(1 + y')^{1/2} = y''.$
   (j) $y' = \sin y.$

2. Find the general solution of the following differential equations.

   (a) $(x \ln x)y' = y, \quad x > 0$
   (b) $axy' = by, \quad a \neq 0$
   (c) $y^2y' + x^2 = 0$
   (d) $y' + ay + b = 0, \quad a \neq 0$
   (e) $y' + 2y \tan 2x = 0$
   (f) $(x^2 + x + 1)y' + (6x + 3)y = 0$
   (g) $y' = e^{x+y} + x^2e^y$
   (h) $yy' = xe^{-x}\sqrt{1-y^2}$
   (i) $y' = 1 + 2x + y + 2xy$
   (j) $x(e^{4y} - 1)y' + (x^2 - 1)e^{2y} = 0, \quad x > 0$
   (k) $ydx - 3xdy = 3xydx$

3. Find the general solution of following first-order linear differential equations.

   (a) $xy' + y = \sin x.$
   (b) $y' + xy = 2x.$
   (c) $xy' = y + (x+1)^2.$
   (d) $y' - 2y = \cos 3x.$
   (e) $\dfrac{dx}{dt} - 2x = t^2e^{2t}.$
   (f) $\dfrac{ds}{du} + s = ue^{-u} + 1.$
   (g) $y' - 3y = \sin 2x.$
   (h) $y' + 4y = 2x + 4x^2.$
   (i) $xy' + y = x^3 + x.$
   (j) $y' + 3y = e^{2x} + 6.$
   (k) $y' = (y+1)\tan x.$
   (l) $xy' + 2(1+x^2)y = 6.$

Final Answers:

1. (a) Order = 2, Degree = 1, Linear.
   (b) Order = 2, Degree = 1, Linear.
   (c) Order = 1, Degree = 2, Non-linear.
   (d) Order = 1, Degree = 1, Non-linear.
   (e) Order = 1, Degree = 2, Non-linear.
   (f) Order = 2, Degree = 1, Non-linear.
   (g) Order = 2, Degree = 2, Non-linear.
   (h) Order = 2, Degree = 1, Linear.
   (i) Order = 2, Degree = 2, Non-linear.
   (j) Order = 1, Degree = 1, Non-linear.

2. (a) $y = C \ln x$
   (b) $y = Cx^{\frac{b}{a}}$
   (c) $y^3 = C - x^3$
   (d) $y = Ce^{-ax} - \frac{b}{a}$
   (e) $y = C \cos 2x$
   (f) $y = C(x^2 + x + 1)^{-3}$
   (g) $y = \ln\left(\frac{1}{C - e^x - x^3/3}\right)$
   (h) $\sqrt{1 - y^2} = (x + 1)e^{-x} + C$
   (i) $y = -1 + Ce^{x+x^2}$
   (j) $e^{2y} + e^{-2y} = -x^2 + 2 \ln |x| + C$
   (k) $y = C\, x^{1/3} e^{-x}$

3. (a) $I.F. = x \qquad y = \frac{-\cos x + C}{x}$
   (b) $I.F. = e^{\int x dx} = e^{x^2/2} \qquad y = 2 + Ce^{-x^2/2}$
   (c) $I.F. = \frac{1}{x} \qquad y = x^2 + x + Cx.$
   (d) $I.F. = e^{-2x} \qquad y = \frac{e^{-2x}}{13}(2 \sin 3x - 3 \cos 3x) + Ce^{2x}$
   (e) $I.F. = e^{-2t} \qquad x = \frac{t^3}{3} e^{2t} + Ce^{2t}$
   (f) $I.F. = e^u \qquad s = \frac{u^2}{2} e^{-u} + 1 + Ce^{-u}$
   (g) $y = \frac{1}{13}(3 \sin 2x - 2 \cos 2x) + Ce^{3x}$
   (h) $y = \frac{1}{2}x^2 + x - \frac{1}{2} + Ce^{-4x}$
   (i) $y = \frac{1}{4}x^3 + \frac{1}{2}x + \frac{C}{x}$
   (j) $y = \frac{1}{2}e^{2x} + 2 + Ce^{-3x}$
   (k) $y + 1 = C \sec x$
   (l) $y = \frac{6 + C}{1 + x^2}$